

Chapter 10

Modules and Finitely Generated Abelian Groups

We now take a detour from the purely linear-algebraic material to study modules—a generalization of vector spaces where the scalars come from a ring rather than a field. The payoff is immediate: the classification of finitely generated abelian groups reduces to “linear algebra over $\mathbb{Z}$.”

Recall from **Lecture 2** that every abelian group $(G, +)$ is a $\mathbb{Z}$ -module: the action $\mathbb{Z} \times G \rightarrow G$ sends $(n, g) \mapsto ng$ (where $ng = g + g + \cdots + g$ for $n > 0$, $0 \cdot g = 0$, and $ng = -(n|g)$ for $n < 0$). This observation is the bridge between group theory and module theory.

10.1 Modules over a Ring

10.1.1 Definitions and Examples

Definition 10.1.1: Module

Let R be a ring (with unity 1). A (left) **R -module** is an abelian group $(M, +)$ equipped with a scalar multiplication $R \times M \rightarrow M$, $(r, m) \mapsto r \cdot m$, satisfying:

- (i) $r(m_1 + m_2) = rm_1 + rm_2$ for all $r \in R$, $m_1, m_2 \in M$.
- (ii) $(r_1 + r_2)m = r_1m + r_2m$ for all $r_1, r_2 \in R$, $m \in M$.
- (iii) $(r_1r_2)m = r_1(r_2m)$ for all $r_1, r_2 \in R$, $m \in M$.
- (iv) $1 \cdot m = m$ for all $m \in M$.

Example 10.1.1 (Vector spaces)

If $R = k$ is a field, then a k -module is the same thing as a vector space over k . So modules generalize vector spaces by allowing the scalars to come from a ring.

Example 10.1.2 (Abelian groups as $\mathbb{Z}$ -modules)

Every abelian group $(G, +)$ is a $\mathbb{Z}$ -module via $n \cdot g = ng$. Conversely, every $\mathbb{Z}$ -module is an abelian group (just forget the $\mathbb{Z}$ -action, which is determined by the group structure anyway). The categories of abelian groups and $\mathbb{Z}$ -modules are the same.

Example 10.1.3 (The ring as a module over itself)

R is an R -module via left multiplication: $r \cdot s = rs$. This is a free module of rank 1.

Example 10.1.4 (R -modules from ring homomorphisms)

If $\varphi : R \rightarrow S$ is a ring homomorphism, then any S -module M becomes an R -module via $r \cdot m = \varphi(r)m$. This is called restriction of scalars.

Example 10.1.5 (Polynomial modules)

Let $R = k[x]$ and V a vector space over k . Any linear operator $T : V \rightarrow V$ makes V into a $k[x]$ -module via $p(x) \cdot v = p(T)(v)$. Different operators give different module structures on the same underlying vector space.

10.1.2 Submodules and Ideals

Definition 10.1.2: Submodule

A **submodule** of an R -module M is a subset $N \subseteq M$ that is itself an R -module under the inherited operations. Equivalently, N is a subgroup of $(M, +)$ that is closed under scalar multiplication: $rn \in N$ for all $r \in R, n \in N$.

Definition 10.1.3: Ideal

A submodule of R (viewed as a module over itself) is called an **ideal**. That is, an ideal $I \subseteq R$ is a subset satisfying:

- (i) I is an abelian subgroup of $(R, +)$.
- (ii) $R \cdot I \subseteq I$: for all $r \in R$ and $a \in I$, we have $ra \in I$.

Example 10.1.6 (Ideals in $\mathbb{Z}$ and $k[x]$)

The ideals of $\mathbb{Z}$ are exactly the subgroups $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for $n \geq 0$. Similarly, the ideals of $k[x]$ are $p(x) \cdot k[x] = \{p(x)q(x) : q(x) \in k[x]\}$. In both cases, every ideal is generated by a single element. Rings with this property are called **principal ideal domains** (PIDs). This is closely related to the Euclidean division algorithm: $\text{span}(p, q) = \text{span}(\text{gcd}(p, q))$.

Note:-

The quotient of an R -module M by a submodule N is again an R -module, with $r \cdot (m + N) = rm + N$. For example, $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}/n$ as a $\mathbb{Z}$ -module, and $k[x]/(p(x)) \cdot k[x]$ is a $k[x]$ -module. The quotient of R by an ideal is not just an R -module but a ring in its own right.

10.1.3 General Facts about Modules: Nothing Is True

Over a field, every vector space has a basis, every linearly independent set extends to one, and every spanning set contains one. Over a general ring, all of these fail.

Example 10.1.7 (Bases need not exist)

The $\mathbb{Z}$ -module $M = \mathbb{Z}/n$ (for $n \geq 2$) has no basis. Any map $\varphi : \mathbb{Z} \rightarrow M$ sending $1 \mapsto m$ cannot be injective (since $nm = 0$ in M but $n \neq 0$ in $\mathbb{Z}$). So no single-element set is linearly independent, and no basis can exist.

Example 10.1.8 (Linearly independent sets may not be part of a basis)

Consider $M = \mathbb{Z}$ as a $\mathbb{Z}$ -module. The element $2 \in \mathbb{Z}$ is linearly independent (if $a \cdot 2 = 0$ then $a = 0$). But $\{2\}$ cannot be extended to a basis of $\mathbb{Z}$: the only bases of $\mathbb{Z}$ are $\{1\}$ and $\{-1\}$, and no set containing 2 is a basis.

Example 10.1.9 (Spanning sets need not contain a basis)

The $\mathbb{Z}$ -module $M = \mathbb{Z}$ is spanned by $\{4, 5\}$ (since $\gcd(4, 5) = 1$, so $1 = 5 \cdot 5 - 6 \cdot 4 \in \text{span}(4, 5)$). But $\{4, 5\}$ is not linearly independent ($5 \cdot 4 - 4 \cdot 5 = 0$), and neither subset $\{4\}$ or $\{5\}$ spans $\mathbb{Z}$. So the spanning set contains no basis.

Example 10.1.10 (Submodules of finitely generated modules need not be finitely generated)

Let $R = k[x_1, x_2, \dots]$ (polynomials in infinitely many variables) and $M = R$ as an R -module. Then M is generated by $\{1\}$. But the ideal $M' = \{f \in R : f(0) = 0\}$ (polynomials with zero constant term) is a submodule not finitely generated: any finite subset only involves finitely many variables x_i , so cannot span x_j for large j .

By contrast, this pathology does not occur over Noetherian rings (including $\mathbb{Z}$ and $k[x_1, \dots, x_n]$).

10.1.4 Module Homomorphisms

Definition 10.1.4: Module homomorphism

Let M, N be R -modules. A map $\varphi \in \text{Hom}_R(M, N)$ is a module homomorphism if

$$\varphi(u + w) = \varphi(u) + \varphi(w) \quad \text{and} \quad \varphi(av) = a\varphi(v)$$

for all $u, w \in M, a \in R$.

Note:-

$\text{Hom}_R(M, N)$ is itself an R -module: $(\varphi + \psi)(v) = \varphi(v) + \psi(v)$ and $(a\varphi)(v) = a\varphi(v)$. For free modules, things work as expected: $\text{Hom}_R(R^m, R^n) \cong R^{m \times n}$ (determined by images of basis vectors, just like linear maps between vector spaces). But for non-free modules, $\text{Hom}_R(M, N)$ can vanish even when both M and N are nonzero. For instance, $\text{Hom}_R(k, k[x]) = 0$ when $R = k[x]$ and $M = k = k[x]/(x)$: any φ must satisfy $x \cdot \varphi(1) = \varphi(x \cdot 1) = \varphi(0) = 0$, so $\varphi(1)$ is killed by x in $k[x]$, forcing $\varphi(1) = 0$.

10.1.5 Free Modules and Direct Sums

Definition 10.1.5: Free module

An R -module M is free if it has a basis: a set $\{e_i\}_{i \in I}$ such that every element of M can be written uniquely as a finite R -linear combination $\sum r_i e_i$. Equivalently, $M \cong R^{(I)} = \bigoplus_{i \in I} R$. The rank of a free module is $|I|$ (well-defined when R is commutative).

Note:-

Over a field, every module (= vector space) is free. Over $\mathbb{Z}$, free modules are the groups $\mathbb{Z}^n$. The module $\mathbb{Z}/n$ is not free for $n \geq 2$.

Definition 10.1.6: Direct sum of modules

The direct sum $M_1 \oplus M_2$ is the set of pairs (m_1, m_2) with componentwise operations. More generally, $\bigoplus_{i \in I} M_i$ consists of tuples $(m_i)_{i \in I}$ with only finitely many nonzero entries. This has the universal property: maps out of $\bigoplus M_i$ correspond to families of maps out of each M_i .

Definition 10.1.7: Exact sequences of modules

A sequence of R -module homomorphisms

$$\cdots \rightarrow M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \rightarrow \cdots$$

is exact at M_i if $\text{Im}(f_{i-1}) = \text{Ker}(f_i)$. A short exact sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ means f is injective, g is surjective, and $\text{Im}(f) = \text{Ker}(g)$, i.e., $C \cong B/f(A)$. A short exact sequence splits if $B \cong A \oplus C$. Over a PID, every short exact sequence of finitely generated modules with C free splits (analogous to Maschke's theorem).

10.2 Classification of Finitely Generated Abelian Groups

The main theorem of this lecture is:

Theorem 10.2.1 Classification of finitely generated abelian groups

Every finitely generated abelian group G is isomorphic to a product of cyclic groups:

$$G \cong \mathbb{Z}/n_1 \times \mathbb{Z}/n_2 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$$

where $n_i \geq 2$ and $\ell \geq 0$. Moreover, using the Chinese Remainder Theorem ($\mathbb{Z}/mn \cong \mathbb{Z}/m \times \mathbb{Z}/n$ when $\gcd(m, n) = 1$), we can arrange each n_i to be a prime power. In this form, the decomposition is unique up to reordering.

The proof reduces to “linear algebra over $\mathbb{Z}$,” following Artin §14.4–14.7. The strategy has two ingredients: first, express any finitely generated $\mathbb{Z}$ -module as $\mathbb{Z}^n / \text{Im}(T)$ for some $T \in \text{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$; second, diagonalize T by changes of basis over $\mathbb{Z}$.

10.2.1 Step 1: Presenting a Module as a Quotient

Proposition 10.2.1 Presentation of finitely generated $\mathbb{Z}$ -modules

If M is a finitely generated $\mathbb{Z}$ -module, then there exist $m, n \geq 0$ and $T \in \text{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$ such that

$$M \cong \mathbb{Z}^n / \text{Im}(T).$$

Equivalently, there is an exact sequence $\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \rightarrow M \rightarrow 0$.

This says: quotient by a $\mathbb{Z}$ -submodule of $\mathbb{Z}^n$ is the same as quotient of an abelian group by a subgroup. We need a lemma:

Lemma 10.2.1 Submodules of $\mathbb{Z}^n$ are free

Any submodule of $\mathbb{Z}^n$ is finitely generated and free of rank $\leq n$.

Proof: By induction on n . For $n = 1$: subgroups of $(\mathbb{Z}, +)$ are $\{0\}$ or $\mathbb{Z}a$ for some $a \in \mathbb{Z} \setminus \{0\}$, both free of rank ≤ 1 .

Assume the result for $\mathbb{Z}^{n-1}$, and let $N \subseteq \mathbb{Z}^n$ be a submodule. The projection $\pi : \mathbb{Z}^n \rightarrow \mathbb{Z}^{n-1}$ sending $(a_1, \dots, a_n) \mapsto (a_1, \dots, a_{n-1})$ restricts to a homomorphism $\pi|_N : N \rightarrow \mathbb{Z}^{n-1}$.

The image $\text{Im}(\pi|_N)$ is a submodule of $\mathbb{Z}^{n-1}$, hence free and finitely generated by induction. The kernel $\text{Ker}(\pi|_N) = N \cap (\mathbb{Z} \times 0 \times \dots \times 0)$ is a subgroup of $\mathbb{Z}$, hence free of rank 0 or 1.

If $\text{Ker}(\pi|_N)$ and $\text{Im}(\pi|_N)$ are both finitely generated (and free), then so is N . Let $e_1, \dots, e_k$ be generators of $\text{Ker}(\pi|_N)$ (a basis) and $g_1 = \pi(f_1), \dots, g_m = \pi(f_m)$ generators of $\text{Im}(\pi|_N)$. For any $x \in N$, write $\pi(x) = \sum a_i g_i$, so $\pi(x - \sum a_i f_i) = 0$, meaning $x - \sum a_i f_i \in \text{Ker}(\pi|_N) = \text{span}(e_1, \dots, e_k)$. Thus $x \in \text{span}(e_1, \dots, e_k, f_1, \dots, f_m)$, so $\{e_i, f_j\}$ generates N . That this set is a basis (i.e., free) follows from the directness of $\text{Ker} \cap \text{Im} = 0$. $\square$

Proof of the proposition: Let M be finitely generated with generators $e_1, \dots, e_n$. Define $\varphi : \mathbb{Z}^n \rightarrow M$ by $(a_1, \dots, a_n) \mapsto \sum a_i e_i$. This is surjective. The kernel $N = \text{Ker}(\varphi) \subseteq \mathbb{Z}^n$ is a submodule, hence finitely generated by the lemma. Let $f_1, \dots, f_m$ be generators of N . Define $T : \mathbb{Z}^m \rightarrow \mathbb{Z}^n$ by $T(b_i) = f_i$. Then $\text{Ker}(\varphi) = \text{Im}(T)$, giving the exact sequence

$$\mathbb{Z}^m \xrightarrow{T} \mathbb{Z}^n \xrightarrow{\varphi} M \rightarrow 0$$

and thus $M \cong \mathbb{Z}^n / \text{Im}(T)$. $\square$

10.2.2 Step 2: Diagonalization over $\mathbb{Z}$ (Smith Normal Form)

The key difference from linear algebra over a field: we cannot divide freely in $\mathbb{Z}$. The substitute is the Euclidean algorithm, which governs what “row and column operations” are available.

Definition 10.2.1: Divisibility

The **divisibility** of a nonzero element $x = (a_1, \dots, a_n) \in \mathbb{Z}^n$ is the largest $d \in \mathbb{Z}_+$ such that $x = dy$ for some $y \in \mathbb{Z}^n$. Equivalently, $d = \gcd(a_1, \dots, a_n)$. An element is **primitive** if its divisibility is 1.

Lemma 10.2.2 Primitive elements and bases

An element of $\mathbb{Z}^n$ can be part of a basis if and only if it is primitive. More generally, $v \in \mathbb{Z}^n$ with divisibility d satisfies $v = d \cdot w$ where w is a basis element.

Proof: Elements of a basis $(e_1, \dots, e_n)$ are primitive: linear independence prevents $e_i = d(\sum a_j e_j)$ for $d > 1$.

Conversely, let $v = a_1 e_1 + \dots + a_n e_n$ be primitive. Without loss of generality, $a_1 \neq 0$ and $|a_1| = \min\{|a_i| : a_i \neq 0\}$. For each $k \geq 2$, write $a_k = q_k a_1 + r_k$ by Euclidean division. Change basis to $(e'_1 = e_1 + \sum_{k \geq 2} q_k e_k, e_2, \dots, e_n)$. In this basis, $v = a_1 e'_1 + r_2 e_2 + \dots + r_n e_n$ with $0 \leq r_k < |a_1|$.

Repeating this process (which strictly decreases the minimum nonzero coefficient at each step) terminates in finitely many steps with $v = d \cdot e''_1$ for some basis vector e''_1 . Since v is primitive, $d = 1$. $\square$

Theorem 10.2.2 Smith Normal Form

For any $T \in \text{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$, there exist bases $(e_1, \dots, e_m)$ of $\mathbb{Z}^m$ and $(f_1, \dots, f_n)$ of $\mathbb{Z}^n$, and positive integers $d_1, \dots, d_r$ (where $r \leq \min(m, n)$ is the rank of T), such that

$$T(e_i) = \begin{cases} d_i f_i & \text{if } 1 \leq i \leq r \\ 0 & \text{if } i > r. \end{cases}$$

In matrix form:

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & & & 0 \\ & d_2 & & \\ & & \ddots & \\ 0 & & & d_r & \\ & & & & 0 \end{pmatrix}.$$

Moreover, the construction gives $d_1 = \min\{\text{div}(T(x)) : x \notin \text{Ker}(T)\} = \text{gcd of all entries of } \mathcal{M}(T)$.

Proof: By induction on m .

If $T = 0$, the statement is trivial for any m, n .

Case $m = 1$: T is determined by $T(1) \in \mathbb{Z}^n$. Let $d = \text{div}(T(1))$. By the lemma, $T(1) = df_1$ where f_1 can be extended to a basis of $\mathbb{Z}^n$. Take $e_1 = 1$.

Inductive step: Assume $T \neq 0$. Let $d_1 = \min\{\text{div}(T(x)) : x \notin \text{Ker}(T)\}$, and choose e_1 with $\text{div}(T(e_1)) = d_1$. Since $T(e_1)$ has divisibility d_1 , write $T(e_1) = d_1 f_1$ with f_1 primitive, and extend f_1 to a basis $(f_1, \dots, f_n)$ of $\mathbb{Z}^n$. Extend e_1 to a basis $(e_1, \dots, e_m)$ of $\mathbb{Z}^m$ (using the lemma, since e_1 is primitive: if $e_1 = d \cdot e'_1$ then $\text{div}(T(e'_1)) = d_1/d$, contradicting minimality unless $d = 1$).

In the bases (e_i) and (f_j) , the matrix of T has the form

$$\mathcal{M}(T) = \begin{pmatrix} d_1 & a_2 & \cdots & a_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}$$

where T' is the restriction to $\text{span}(e_2, \dots, e_m)$ composed with projection to $\text{span}(f_2, \dots, f_n)$.

For $j \geq 2$, write $a_j = q_j d_1 + r_j$ with $0 \leq r_j < d_1$. Change basis of $\mathbb{Z}^m$ to $(e_1, e'_2 = e_2 - q_2 e_1, \dots, e'_m = e_m - q_m e_1)$. The matrix becomes

$$\begin{pmatrix} d_1 & r_2 & \cdots & r_m \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

If any $r_j \neq 0$, then $\text{div}(T(e'_j)) \leq r_j < d_1$, contradicting the minimality of d_1 . So $r_j = 0$ for all $j \geq 2$, and the matrix is

$$\begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & \mathcal{M}(T') & \\ 0 & & & \end{pmatrix}.$$

By induction on m , we can diagonalize T' by further basis changes in $\text{span}(e_2, \dots, e_m)$ and $\text{span}(f_2, \dots, f_n)$, giving the desired diagonal form. $\square$

Note:-

The Smith Normal Form is the integer analogue of row reduction. Over a field, Gaussian elimination diagonalizes any matrix (in fact reduces it to have 1's and 0's on the diagonal). Over $\mathbb{Z}$, the Euclidean algorithm plays the role of division, and the diagonal entries d_i need not be 1. The construction actually forces $d_1 \mid d_2 \mid \cdots \mid d_r$ (the invariant factor divisibility chain), though we do not prove this here.

10.2.3 Step 3: Completing the Classification

Proof of the classification theorem: By Proposition 1 (presentation as quotient), any finitely generated abelian group M satisfies $M \cong \mathbb{Z}^n / \text{Im}(T)$ for some $T \in \text{Hom}(\mathbb{Z}^m, \mathbb{Z}^n)$. By the Smith Normal Form,

after choosing appropriate bases, $\text{Im}(T)$ is spanned by $d_1 f_1, \dots, d_r f_r$ for some positive integers $d_i > 0$ and $r \leq n$. Therefore

$$M \cong \mathbb{Z}^n / \text{span}(d_1 f_1, \dots, d_r f_r) \cong \mathbb{Z}/d_1 \times \cdots \times \mathbb{Z}/d_r \times \mathbb{Z}^{n-r}.$$

The factors $\mathbb{Z}/d_i$ with $d_i = 1$ are trivial and can be dropped. Relabeling the remaining $d_i \geq 2$ as $n_1, \dots, n_k$ and setting $\ell = n - r$ gives the decomposition

$$M \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell.$$

By the Chinese Remainder Theorem, each $\mathbb{Z}/n_i$ with $n_i = p_1^{a_1} \cdots p_s^{a_s}$ decomposes as $\mathbb{Z}/p_1^{a_1} \times \cdots \times \mathbb{Z}/p_s^{a_s}$. Rearranging gives the prime power form. $\square$

Example 10.2.1 (Finitely generated abelian groups of small order)

- Order 4: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. These are not isomorphic (the first has an element of order 4, the second does not).
- Order 6: $\mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/3$ (only one, since $\gcd(2, 3) = 1$).
- Order 8: $\mathbb{Z}/8$, $\mathbb{Z}/4 \times \mathbb{Z}/2$, or $\mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/2$.
- Order 12: $\mathbb{Z}/12 \cong \mathbb{Z}/4 \times \mathbb{Z}/3$, or $\mathbb{Z}/2 \times \mathbb{Z}/6 \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3$.

Note:-

The number ℓ is the **free rank** (or **Betti number**) of M , and the integers $n_1, \dots, n_k$ (in prime power form) are the **elementary divisors**. These invariants completely determine M up to isomorphism. The free rank counts the number of “copies of $\mathbb{Z}$,” while the elementary divisors encode the torsion (elements of finite order).

Exercise 10.2.1 Modules and finitely generated abelian groups exercises

- (1) Classify all abelian groups of order 36 up to isomorphism.
- (2) Let $T : \mathbb{Z}^2 \rightarrow \mathbb{Z}^2$ be the linear map with matrix $\begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. Compute the Smith Normal Form of T and determine the group $\mathbb{Z}^2 / \text{Im}(T)$.
- (3) Prove that a finitely generated abelian group is free (i.e., isomorphic to $\mathbb{Z}^n$ for some n) if and only if it is torsion-free (i.e., the only element of finite order is 0).
- (4) Let G be a finitely generated abelian group with $|G| = p^n$ for a prime p . Show that $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ where $a_1 + \cdots + a_k = n$ and $a_i \geq 1$. How many such groups are there (as a function of the partition structure of n)?
- (5) Show that $\mathbb{Z}/m \otimes_{\mathbb{Z}} \mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$. (Hint: use the presentation $\mathbb{Z}/m = \mathbb{Z}/m\mathbb{Z}$ and the right-exactness of the tensor product.)

Solution

(1) We have $36 = 2^2 \cdot 3^2$. For the 2-part: $\mathbb{Z}/4$ or $\mathbb{Z}/2 \times \mathbb{Z}/2$. For the 3-part: $\mathbb{Z}/9$ or $\mathbb{Z}/3 \times \mathbb{Z}/3$. Taking all combinations:

$$\mathbb{Z}/4 \times \mathbb{Z}/9 \cong \mathbb{Z}/36, \quad \mathbb{Z}/4 \times \mathbb{Z}/3 \times \mathbb{Z}/3, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/9, \quad \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/3 \times \mathbb{Z}/3.$$

So there are exactly 4 abelian groups of order 36.

(2) Start with $A = \begin{pmatrix} 6 & 4 \\ 3 & 5 \end{pmatrix}$. The gcd of all entries is 1, but let us apply row/column operations over $\mathbb{Z}$.

$$\begin{aligned} R_1 &\rightarrow R_1 - 2R_2: \begin{pmatrix} 0 & -6 \\ 3 & 5 \end{pmatrix}. \text{ Swap rows: } \begin{pmatrix} 3 & 5 \\ 0 & -6 \end{pmatrix}. \quad C_2 \rightarrow C_2 - C_1: \begin{pmatrix} 3 & 2 \\ 0 & -6 \end{pmatrix}. \quad C_1 \rightarrow C_1 - C_2: \\ &\begin{pmatrix} 1 & 2 \\ 6 & -6 \end{pmatrix}. \quad C_2 \rightarrow C_2 - 2C_1: \begin{pmatrix} 1 & 0 \\ 6 & -18 \end{pmatrix}. \quad R_2 \rightarrow R_2 - 6R_1: \begin{pmatrix} 1 & 0 \\ 0 & -18 \end{pmatrix}. \end{aligned}$$

Smith Normal Form: $\text{diag}(1, 18)$. So $\mathbb{Z}^2/\text{Im}(T) \cong \mathbb{Z}/1 \times \mathbb{Z}/18 \cong \mathbb{Z}/18 \cong \mathbb{Z}/2 \times \mathbb{Z}/9$.

Check: $\det(A) = 30 - 12 = 18$, and $|\mathbb{Z}^2/\text{Im}(T)| = |\det(T)| = 18$. Consistent.

(3) One direction: if $G \cong \mathbb{Z}^n$, then G is torsion-free (only $0 \in \mathbb{Z}^n$ has finite order).

Conversely, if G is finitely generated and torsion-free, write $G \cong \mathbb{Z}/n_1 \times \cdots \times \mathbb{Z}/n_k \times \mathbb{Z}^\ell$ by the classification theorem. Each $\mathbb{Z}/n_i$ contributes a nonzero element of finite order (namely $\bar{1}$, which has order n_i). Since G is torsion-free, $k = 0$, so $G \cong \mathbb{Z}^\ell$.

(4) Since $|G| = p^n$, the classification gives $G \cong \mathbb{Z}/p^{a_1} \times \cdots \times \mathbb{Z}/p^{a_k}$ with $a_i \geq 1$ and $p^{a_1} \cdots p^{a_k} = p^n$, i.e., $a_1 + \cdots + a_k = n$. The number of such groups equals the number of partitions of n . For example, p^4 gives partitions $(4), (3, 1), (2, 2), (2, 1, 1), (1, 1, 1, 1)$, so 5 groups.

(5) From the short exact sequence $0 \rightarrow \mathbb{Z} \xrightarrow{\times m} \mathbb{Z} \rightarrow \mathbb{Z}/m \rightarrow 0$, tensor with $\mathbb{Z}/n$. Right-exactness gives:

$$\mathbb{Z}/n \xrightarrow{\times m} \mathbb{Z}/n \rightarrow \mathbb{Z}/m \otimes \mathbb{Z}/n \rightarrow 0.$$

The map $\times m : \mathbb{Z}/n \rightarrow \mathbb{Z}/n$ has image $m\mathbb{Z}/n \cong \mathbb{Z}/\gcd(m, n)$ (the image of multiplication by m in $\mathbb{Z}/n$ is the cyclic subgroup of order $n/\gcd(m, n)$). So:

$$\mathbb{Z}/m \otimes \mathbb{Z}/n \cong (\mathbb{Z}/n)/(m\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$